MIL-F-46470D (MU)
17 December 1972
SUPERSEDING
MIL-F-46470C (MU)
15 August 1967

MILITARY SPECIFICATION

FUZES, PD, M525 AND M527B1, HEAD ASSEMBLY FOR

1. SCOPE

1.1 This specification covers the Head Assembly for the Fuzes, PD, M525 and M527B1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance for Ordnance
MIL-P-116 - Preservation, Methods of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105D)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1390

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DRAWINGS

ARMY

8798660 - Plug, Slider
8798661 - Spring, Slider
8798662 - Body Assembly
8798677 - Slider
8800197 - Fuze, PD, M525
8800199 - Complete Head Assembly
9221314 - Box, Fiber for Head Assembly for Fuze, PD, M525 or M525A2E1
9221315 - Box, Packing, Ammunition for Head Assembly for Fuze, PD, M525 or M525A2E1

PUBLICATIONS

ET-8800199 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Material. - Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Metal Parts. - The metal parts shall comply with all requirements specified on Drawing (dwg.) 8800199, and with all requirements specified in applicable specifications, except as modified herein.

3.3 Complete Head Assembly

3.3.1 Transportation Vibration. - The head assembly shall withstand the test specified in MIL-STD-331 when tested as specified in 4.4.1.

3.3.2 Positioning. - The firing pin shall move and shoulder on the slider when a dead load of 50 pounds, maximum (max.) is applied to the striker assembly when tested as specified in 4.4.2.

3.3.3 Jolt and Jumble. - The head assembly shall withstand the tests specified in MIL-STD-331 when tested as specified in 4.4.3.

3.3.4 Static Functioning. - The head assembly shall comply with the requirements specified in 3.4.

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3.3.5 Ballistic Performance

3.3.5.1 Non-Functioning. - The head assembly shall not function when tested as specified in 4.4.4.1.

3.3.5.2 Functioning. - There shall be no evidence of a premature burst and the head assembly shall cause the shell to function when tested as specified in 4.4.4.2 (see 6.7).

3.4 Movement Assembly

3.4.1 Arming Delay. - The movement assembly shall arm in not less than 2.0 or more than 4.0 seconds when tested as specified in 4.4.5.

3.4.2 Arming. - The movement assembly shall arm as evidenced by retraction of the firing pin assembly, after being subjected to an acceleration of 300 minus 5 gravities applied in the direction simulating ballistic firing when tested as specified in 4.4.6.

3.4.3 Non-Arming. - The movement assembly shall not arm when subjected to an acceleration of 150 plus 5 gravities applied in the direction simulating ballistic firing when tested as specified in 4.4.7.

3.4.4 Non-Positioning. - The firing pin shall not move when a dead load of 10 plus 1 pounds is applied to the flat end when tested as specified in 4.4.8.

3.5 Embrittlement of Springs (Spring, Plunger; Spring, Release; Spring, Setback Detent and Spring, Balance Lock Pin). - The springs shall not break or crack when extended to the point of minimum permanent distortion when tested as specified in 4.4.9.

3.6 Assembly of Firing Pin. - The load required to assemble the firing pin into the component parts of the firing pin assembly shall not exceed 40 pounds, max.

3.7 First Article Inspection. - This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in 4.2.

3.8 Workmanship. - All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the

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cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 General Quality Assurance Provisions. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of Product. - At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government Verification. - Using the contractor's written quality assurance procedure, this detailed specification, with applicable drawings and other contractual documents as a guide, the Government inspector shall verify

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at scheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "Critical" be accepted solely on the basis of the contractor's records.

4.2 First Article Inspection

4.2.1 Submission. - The contractor shall submit a first article sample as designated by the Contracting Officer (see 6.6) for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 362 complete head assemblies and 25 complete sets of parts (i.e., 25 each of every component and every subassembly) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be Performed. - Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in this specification and any or all requirements of the applicable drawings. In addition, of the 362 head assemblies, 112 will be subjected to the following tests in the order specified. Quantities for each test are as indicated. In the event the entire initial production quantity passes the inspections specified, the remaining 250 head assemblies will be retained by the engineering agency as a control sample.

a. Transportation Vibration (all 112 sample units) - (see 4.4.1)
b. Non-Arming (all 112 sample units) - (see 4.4.7)
c. Positioning (all 112 sample units) - (see 4.4.2)
d. Jolt and Jumble (12 sample units after conducting a, b and c above) - (see 4.4.3)
e. Arming (50 sample units after conducting a, b and c above) - (see 4.4.6)
f. Arming Delay (50 sample units after conducting a, b and c above) - (see 4.4.5)

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4.2.3 Ballistic Test. - The contractor shall submit 1,000 consecutively produced complete head assemblies which have been produced by the same production process as those submitted in accordance with 4.4.4.1 or 4.4.4.2, as applicable, except that quantities and test conditions shall be as specified in Table I. Temperature variation may be plus or minus 5°F.

TABLE I

Requirements	Quantity	Firing Temperature	Charge
Non-Functioning (see 3.3.5.1)	250	70°F	0
Functioning (see 3.3.5.2)	125	125°F	0
Functioning	125	125°F	4
Functioning	125	70°F	0
Functioning	125	70°F	4
Functioning	125	- 40°F	0
Functioning	125	- 40°F	4

4.2.4 Rejection. - If any assembly, subassembly or component part fails to comply with the requirements for the inspections performed in accordance with 4.2.2, the first article production quantity shall be rejected. In the ballistic test, if any round functions in the non-functioning phase, or if more than five (5) rounds fail to function in any group of 125 fired for functioning, or if more than 25 rounds fail in the total of 750 fired in the functioning phase, the first article production quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any or all of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article production sample or portion thereof. Until a first article production sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer.

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4.3 Inspection Provisions for Production

4.3.1 Lot Formation. - The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. Changes to the processes, specifications or drawings, not affecting safety performance, interchangeability or storage, as determined by the Government shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105. Unless otherwise approved by the Contracting Officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for delivery during the contract period. Inspection lots of components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and Government inspector. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination. - Inspection for Critical Defects (and Major Defects when so spacified) shall be 100 percent. Sampling plans and procedures for Major and Minor Defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor Defect and an AQL of 0.40 percent for each Major Defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Workmanship. - Applies to all components and subassemblies not hereinafter specifically listed in 4.3.2.2 through 4.3.2.7 (see 3.8 and applicable drawings).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical: None defined.			

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Major:	AQL 1.0 percent per defect		
101.	Operation missing or improperly performed	Visual	01001
102.	Burr or chip on or at intersection of sliding or other functional surfaces	Visual	01002
103.	Finish on sliding or other functional surface improper	Visual	01003
104.	Component missing or improperly assembled	Visual	01004
105.	Protective finish inadequate	Visual	01005
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship other than that covered under Major category above (see 3.8)	Visual	01006

4.3.2.2 Head (see dwg. 8800201 covering a detail of dwg. 8800199).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Concentricity of movement assembly counterbore with movement assembly cavity	Gage	02001
102.	Pitch diameter of thread	Gage	02002
103.	Major diameter of thread	Gage	02003
Minor:	AQL 4.0 percent		
201.	Concentricity of striker hole with movement assembly cavity	Gage	02004
202.	Diameter of movement assembly cavity, max.	Gage	02005
203.	Depth to striker seat	Gage	02006
204.	Length of thread pilot, max.	Gage	02007
205.	Width of thread undercut, max.	Gage	02008
206.	Finish improper	Visual	02009
207.	Evidence of poor workmanship (see 3.8)	Visual	02010
208.	Protective finish inadequate (when applicable)	Visual	02011

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4.3.2.3 Spring, Plunger (see dwg. 8800202 covering a detail of dwg. 8800199).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Load failure at test height	Gage	03001
Minor:	AQL 2.5 percent		
201.	Load failure at compressed height .	Gage	03002
202.	Diameter of small end, max.	Gage	03003
203.	Diameter of large end, max.	Gage	03004
204.	Evidence of poor workmanship (see 3.8)	Visual	03005
205.	Protective finish inadequate	Visual	03006

4.3.2.4 Spring, Release (see dwg. 8800209 covering a detail of dwg. 8800199).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Load failure at test height	Gage	04001
102.	Fails to work freely over .240 inch diameter cylinder	Gage	04002
Minor:	AQL 1.0 percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	04003
202.	Protective finish inadequate	Visual	04004

4.3.2.5 Firing Pin (see dwg. 8800220 covering a detail of dwg. 8800199).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.65 percent		
101.	Profile and diameter of point	Gage	05001
102.	Length of shank, max.	Gage	05002
103.	Length of point	Gage	05003

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Minor:	AQL 1.0 percent		
201.	Length of shank, min.	Gage	05004
202.	Evidence of poor workmanship (see 3.8)	Visual	05005

4.3.2.6 Complete Head Assembly (see dwg. 8800199).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Pull wire missing or insecure	Visual	06001
2.	Firing pin in armed position	Visual	06002
3.	Housing "D" web broken, cracked or missing	Visual	06003
Major:	AQL 0.65 percent		
101.	Distance from firing pin point to shoulder	Gage	06004
102.	Firing pin point damaged	Visual	06005
103.	Movement assembly loosely or improperly crimped	Visual-Manual	06006
Minor:	AQL 4.0 percent		
201.	Diameter of flange (shoulder), min.	Gage	06007
202.	Distance from flange to bottom of crimp, max.	Gage	06008
203.	Diameter over crimp	Gage	06009
204.	Length of cord between upper and lower clips	Gage	06010
205.	Striker fails to return to original position after being depressed	Visual-Manual	06011
206.	Upper or lower clip missing or insecure	Visual-Manual	06012
207.	Evidence of poor workmanship (see 3.8)	Visual	06013
208.	Marking incomplete or unidentifiable	Visual	06014

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4.3.2.7 Tray, Polyethylene (see dwg. 9221314).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Thickness, min. (to be measured near center of tray as well as near edge)	Gage	07001
102.	Tray cracked or broken	Visual	07002
Minor:	AQL 1.0 percent		
201.	Diameter of holes	Gage	07003
202.	Evidence of poor workmanship (see 3.8)	Visual	07004

4.3.2.8 Fiber Box, Prior to Sealing (see dwg. 9221314).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Plastic protector missing, damaged or improperly assembled	Visual	08001
Minor:	AQL 1.0 percent		
201.	Packing component missing, damaged or improperly assembled	Visual	08002
202.	Any head assembly missing or inverted	Visual	08003

4.3.2.9 Box, Fiber Sealed (see dwg. 9221314).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		

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Minor:	AQL 1.5 percent		
201.	Sealing strip incomplete, torn or badly wrinkled	Visual	09001
202.	Box cut, torn or punctured	Visual	09002
203.	Contents loose	Manual	09003
204.	Marking missing, incorrect or unidentifiable	Visual	09004

4.3.2.10 Bag, Barrier Sealed (see dwg. 9221314).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Bag improperly sealed	Visual	10001
102.	Bag pierced or torn	Visual	10002
Minor:	AQL 0.65 percent		
201.	Marking missing, incorrect or unidentifiable	Visual	10003

4.3.2.11 Wood Packing Box, Sealed (see dwg. 9221315).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Box damaged	Visual	11001
102.	Strapping missing, broken or loose	Visual	11002
Minor:	AQL 1.5 percent		
201.	Marking missing, incorrect or unidentifiable	Visual	11003
202.	Strapping improperly assembled	Visual	11004
203.	Nail protruding	Visual	11005
204.	Contents loose	Visual	11006

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4.3.3 Testing

4.3.3.1 Transportation Vibration (see 3.3.1) - Safety Deficiency - Critical Defect - Code No. 13001; Operability Deficiency - Major Defect - Code No. 14001. - The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter L and AQL 0.40 percent except that no Critical Defects shall be permitted and a decision regarding acceptance shall not be made until after the positioning test has been performed. If any assembly is not safe after the test, it shall be considered a Critical Defect. If any assembly is not operable after the test, it shall be considered a Major Defect. The test shall be performed as specified in 4.4.1.

4.3.3.2 Positioning Test (see 3.3.2) - Critical Defect - Code No. 15001. - The same sample assemblies shall be used for this test as were used in the transportation vibration test. If any assembly fails to comply with the applicable requirement, the lot shall be rejected. This test shall be performed as specified in 4.4.2.

4.3.3.3 Jolt and Jumble (see 3.3.3) - Critical Defect - Code No. 16001. - For each shipping lot, a sample of 80 assemblies shall be selected from each day's production for this test. If one or more samples fail to comply with the applicable requirement, the shipping lot shall be rejected. The test shall be performed as specified in 4.4.3. After the successful testing of 10 consecutive shipping lots, the quantity to be tested shall be reduced to 20 samples per day's production with no failures allowed. The failure of a shipping lot will require the requalification of 10 shipping lots before the sample size may be reduced.

4.3.3.4 Arming Delay Test (see 3.4.1) - Minimum Delay Time - Critical Defect - Code No. 17001; Maximum Delay Time - Major Defect - Code No. 18001. - One hundred (100) percent of each lot shall be subjected to this test. Any assembly failing to comply with the minimum delay time requirement shall be classified as a Critical Defect and removed from the lot. Any assembly failing to comply with the maximum delay time shall be classified as a Major Defect and removed from the lot. This test shall be performed as specified in 4.4.5.

4.3.3.5 Arming Test (see 3.4.2) - Major Defect - Code No. 19001. - The sampling plan for this test shall be in accordance with MIL-STD-105, AQL 0.40 percent. Any assembly in which the firing pin does not retract after having been subjected to the arming acceleration, regardless of cause, shall be classed defective. The test shall be performed as specified in 4.4.6.

4.3.3.6 Non-Arming Test (see 3.4.3) - Critical Defect - Code No. 20001. - This test shall be performed 100 percent. Any assembly which fails to comply with the applicable requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.7.

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4.3.3.7 Non-Positioning (see 3.4.4) - Major Defect - Code No. 21001. - The sampling plan for this test shall be in accordance with MIL-STD-105, AQL 0.40 percent. Any assembly in which the firing pin moves when subjected to the required load shall be classed defective. The test shall be performed as specified in 4.4.8.

4.3.3.8 Embrittlement of Springs (see 3.5 and applicable dwgs.) - Major Defect - Code No. 22001. - A sample of 30 springs each shall be selected from each lot for testing. If one or more springs in any sample fail to comply with the applicable requirement, the lot shall be rejected. Testing shall be performed as specified in 4.4.9.

4.3.3.9 Assembly of Firing Pin (see 3.6). - The contractor shall maintain a positive process control which will prevent the acceptance of any firing pin assembly in which the firing pin cannot be assembled with a maximum load of 40 pounds.

4.3.3.10 Load Tests of Gear and Pinion Assemblies, Escape Wheel and Pinion Assembly and Top Plate Assembly (see dwgs. 8800215, 8800224, 8800227, 8800230 and 8800242) - Major Defect - Code No. 23001. - The sampling plan for each of these tests shall be in accordance with MIL-STD-105, Code Letter H and AQL 0.25 percent. Testing shall be performed as specified in 4.4.10.

4.3.3.11 Heat Seal Test (see dwg. 9221314) - Major Defect - Code No. 24001. - Sampling shall be in accordance with Level S4, MIL-STD-105. Acceptance criteria and test procedure for this test shall be as prescribed in specification MIL-P-116.

4.3.3.12 Ballistic Tests. - The following tests shall be performed as specified in 4.4.4.

4.3.3.12.1 Non-Functioning (see 3.3.5.1) - Critical Defect - Code No. 25001.

4.3.3.12.1.1 Thirty-two (32) head assemblies shall be selected from each lot for this test. If one or more samples fail to comply with the requirement of Paragraph 3.3.5.1, the lot shall be rejected. If a lot of head assemblies fails to comply with the requirement of Paragraph 3.3.5.1, or inspection for Critical Defects reveals one or more defective units, the lot shall be rejected and the sample plan for subsequent lots shall be in accordance with 4.3.3.12.1.2. The sample plan for reworked lots and resubmitted lots that have previously failed inspection for Critical Defects shall be in accordance with Paragraph 4.3.3.12.1.2.

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4.3.3.12.1.2 The sample plan shall be in accordance with the non-functioning phase of the first article inspection ballistic test (Paragraph 4.2.3). Of the total 250 samples ballistically tested per lot, 120 samples shall be subjected to the transportation vibration test per MIL-STD-331 prior to ballistic testing. If one or more samples fail to comply with the requirements of 3.3.5.1, the lot shall be rejected. When three consecutive lots have complied with the requirement of Paragraph 3.3.5.1 and inspection for Critical Defects does not reveal one or more defective units per lot, the sample plan for this test shall be in accordance with Paragraph 4.3.3.12.1.1.

4.3.3.12.2 Functioning (see 3.3.5.2) - Major Defect - Code No. 26001; Premature - Critical Defect - Code No. 27001.

4.3.3.12.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, a sample of 125 head assemblies shall be selected for this test. If four or more assemblies fail to comply with the appiicable requirements for impact functioning, or if one or more premature bursts occur, the lot shall be rejected.

4.3.3.12.2.2 After three consecutive lots have met the criteria of 4.3.3.12.2.1, a sample of 50 head assemblies shall be selected from each lot for this test. If three or more assemblies fail to comply with the applicable requirements, the lot shall be rejected. If two assemblies fail, a second sample of 50 head assemblies shall be selected from the lot and tested. If the combined number of failures in both the first and the second samples is four or more, the lot shall be rejected. If one or more premature bursts occur in either or both sample sizes, the lot shall be rejected.

4.3.4 Inspection Equipment. - The inspection equipment required to perform the examinations and tests prescribed in this section is identified by Equipment Tabulation ET-8800199 (see 6.3).

4.3.4.1 Government Rights to Documentation. - Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and Maintenance. - Supply and maintenance of the equipment listed on the ET shall be in accordance with Specification MIL-I-45607.

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4.3.4.3 Government Use of Contractor's Inspection Equipment. - The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test Methods and Procedures

4.4.1 Transportation Vibration. - The sample unarmed complete head assemblies (with pull wire) shall be tested in accordance with the procedure specified in MIL-STD-331, with the exception that any disassembly to determine acceptability shall not be made until the other tests specified in 4.2.2 or 4.3.3.2 have been completed, as applicable. Observation shall be made to determine whether any assembly has become armed during the test, as evidenced by retraction of the firing pin or is otherwise unsafe in any respect. In this connection, a determination shall be made as to whether any sample unit has actually "timed out" and can be made to arm by applying firm upward finger pressure to the firing pin assembly. Operability shall be determined after performance of the positioning test by arming the assemblies as specified in 4.4.6 and observing for retraction of the firing pin assembly after completion of the timing cycle.

4.4.2 Positioning. - The sample unarmed complete head assemblies shall be assembled to fuze body assemblies conforming to dwg. 8798662 (or modified fuze bodies and sliders in accordance with the ET) complete with empty (unloaded) sliders conforming to dwg. 8798677 and the specified load applied to the striker. Observation of the interior of the slider cavity shall be made with a light source such as an otoscope (or other approved instrument) in order to determine that the firing pin has actually detented the slider and that the firing pin shoulder has positioned against the flat surface of the slider. (NOTE: For the performance of this test on the pre-production quantity, the pull wire does not need to be fastened to the assembly. It was removed before performing the non-arming test.)

4.4.3 Jolt and Jumble. - The sample unarmed complete head assemblies shall be assembled to fuze body assembly, dwg. 8798662; empty (unloaded) sliders, dwg. 8798677; slider plugs, dwg. 8798660 and slider springs, dwg. 8798661. The positioning load shall be applied to the striker and the assemblies then tested in accordance with the procedures specified in MIL-STD-331, except that the jumble test shall be conducted utilizing a jumble fixture similar to that specified in ET-8800199. The assembly shall be considered to have passed the tests if after disassembly of the slider plug and slider spring, the slider is retained in its cavity by the detented firing pin, as determined by observation with an otoscope (or other approved instrument). (NOTE: For the initial production sample, the pull wire must be reassembled since it will have been removed in order to conduct previous tests.)

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4.4.4 Ballistic Performance

4.4.4.1 Non-Functioning. - The sample head assemblies shall be assembled into complete M525 Fuzes as shown on dwg. 8800197 using an approved lot of M525 PD Fuzes (see 6.4). The fuzes shall then be assembled to 60MM, M94A4 HE complete round cartridges with a propellant charge consisting of an ignition cartridge only (Charge 0) and fired horizontally at a 1/4 inch thick plywood target placed 200 feet from the muzzle of the mortar tube. Observation shall be made for functioning on the target.

4.4.4.2 Functioning. - The sample head assemblies shall be assembled into complete M525 PD Fuzes as shown on dwg. 8800197 using an approved lot of M525 PD Fuzes (see 6.4). The fuzes shall then be assembled to 60MM, M94A4 HE complete round cartridges and fired at an elevation of 45 degrees or greater for impact on firm ground. For the first article production ballistic firing, sample quantities shall be as specified in 4.2.3, Table I, and applicable conditioning shall be for 24 hours, minimum. For tests of production lots, one half of the sample quantity shall be fired at ambient temperature with ignition cartridges only (Chare 0) and the remaining sample units shall be fired at ambient temperature with four propellant increments (Charge 4). Observation shall be made for premature bursts and for functioning upon ground impact.

4.4.5 Arming Delay. - The locking detent pin of the unarmed movement assemblies shall be manually or mechanically initiated. The interval between initiation and firing pin arming shall be measured with a suitable timing device and the time recorded, rounded off to the nearest 0.1 second.

4.4.6 Arming. - This test is to be performed upon completely crimped unarmed movement assemblies and shall be accomplished prior to non-arming tests. The unarmed movement assemblies shall be tested as specified in 4.4.7 using the arming acceleration speed. (NOTE: For tests conducted on complete head assemblies, in order to determine operability after the performance of the transportation vibration test, the pull wire must be removed.)

4.4.7 Non-Arming. - The unarmed movement assemblies shall be mounted in a centrifuge with the axis of the firing pin parallel with the rotating arm, and the pointed end of the firing pin away from the axis of rotation. The acceleration shall be measured at the base of the setback sleeve and shall be attained within 10 seconds of rest. After removing the assemblies from the centrifuge, they shall be examined to determine whether the assembly has armed, as evidenced by: (1) The position of the setback detent, (2) The orientation of the slot in the upper bushing with the hole in the mainspring retainer, or (3) The retraction of the firing pin assembly. In the case of initial production tests (conducted on complete head assemblies, less pull wire), firm upward finger pressure shall be applied to the firing pin assembly to determine whether it can be made to retract. If any sample unit can be made to arm in this manner, it shall be considered to have failed the test.

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4.4.8 Non-Positioning. - The sample unarmed movement assemblies shall be placed in a fixture which supports the housing and an initial load of two pounds applied to the flat end of the firing pin in order to take up any "play" within the assembly. An additional dead load of eight pounds shall then be applied and observation made for conformance to the applicable requirement. A dial indicator or other approved instrument shall be used to determine if the firing pin has moved.

4.4.9 Embrittlement of Springs. - The sample springs shall be extended to the minimum point of permanent distortion by any method satisfactory to the Government.

4.4.10 Load Tests. - The sample assemblies shall be subjected to the loads required by the applicable drawings by any method satisfactory to the Government.

4.4.11 Heat Seal. - The sample sealed barrier bags shall be tested in accordance with the procedure specified in Paragraph 4.4.4.3 of MIL-P-116.

5. PREPARATION FOR DELIVERY

5.1 Packaging

5.1.1 Level A. - The head assembly shall be preserved and packaged in accordance with dwg. 9221314.

5.1.2 Level C. - Packaging shall be the same as Level A except that the barrier bag is not required.

5.2 Packing

5.2.1 Level A. - The fuze shall be packed in accordance with dwg. 9221315.

5.2.2 Level C. - Packing shall be the same as Level A except the packing box shall be Grade C and the filler shall be commercial chipboard material.

5.3 Marking. - Marking of unit and exterior containers shall be in accordance with dwgs. 9221314 and 9221315.

6. NOTES

6.1 Ordering Data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.

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6.2 Inspection Code Numbers. - The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection Equipment Designs. - Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawings numbers on the Equipment Lists referenced on the Equipment Tabulation. Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE Designs. - SIE designs may consist of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-110, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor Designs. - Contractor designs are required for all inspection equipment for which SIE designs are not specified, and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment or with SIE designs, if not otherwise covered thereby (see 6.3.1.c.). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

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6.3.3 Submission of Designs for Approval. - Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Approval of M525 PD Fuzes for Use in Ballistic Tests.- Approval of M525 PD Fuzes for use in the ballistic tests prescribed herein shall be determined by selecting a random sample of 300 from a lot of 5000 or more and, after assembly to M49A4 HE complete round cartridges, firing at an angle of 45 degrees elevation or greater for impact on firm ground. One hundred fifty (150) of the sample units shall be fired with ignition cartridges only (Charge 0) and 150 sample units shall be fired with four propellent increments (Charge 4). If there are no premature bursts, and if no more than two duds occur, the lot shall be approved for use.

6.5 Quality Assurance Provisions of Other Documents. - The quality assurance provisions of other documents which are required to be part of the contractor's written description include, but are not limited to, those which relate to plating or other protective coating, physical chemical and other properties, for which specific quality assurance or inspection provisions are included in the applicable material and process specification.

6.6 Submission of First Article Production Samples. - Instructions as to the location for the evaluation of initial production shall be obtained from the Contracting Officer. Upon receipt of such request, the Contracting Officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-F, Dover, New Jersey.

6.7 Premature Bursts. - Premature burst is defined as the functioning of the fuze in the weapon bore or in flight before impact.

Custodian:
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